Alkenes

Alkenes are unsaturated hydrocarbons with atleast one double bonds hence the do not containing the maximum amount of hydrogen.

General formula: C_nH_{2n}

Properties

1–Insoluble in water due to their non–polar nature, but soluble in non–polar organic solvents
2–Flammable and burn with a smoky flame due to higher carbon content relative to alkanes.
3–More reactive than alkanes, undergoing addition reactions (e.g. with bromine, hydrogen, steam) due to the presence of the carbon–carbon double bond (C=C).
4–*States (C_2 to C_4: Gases / C_5 to C_{18}: Liquids / C_{19} and above: Solids) at room temperature.
5–*Boiling/Melting Points increase with chain length due to increased van der Waals forces, but are slightly lower than equivalent alkanes due to reduced molecular symmetry.
6–Viscosity increases with chain length due to stronger intermolecular forces.
7–Typically colourless and can have faint, sweetish or petrol–like odours.

Use

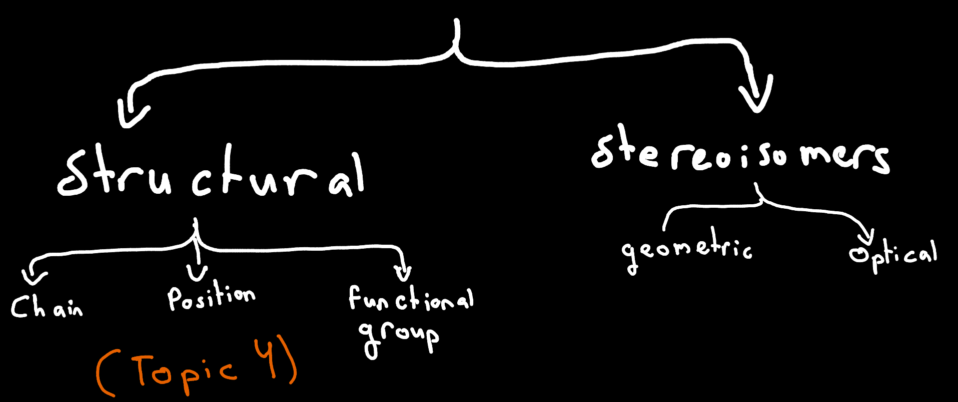
1–Making polymers
2–ripening / maturing fruit e.g. bananas (ethene specific use)

The double bond

A double bond is not two identical bonds. It is a combination of two different types of covalent bonds: one σ bond and one π bond (explained in more details in chapter 3).

Feature	Sigma (σ) Bond	Pi (π) Bond
Formation	Head-on overlap of orbitals	Sideways overlap of p-orbitals
Strength	Strong	Weaker
Rotation	Allows free rotation	Locks atoms, prevents rotation
Electron Location	Directly between nuclei	Above & below the bond axis
Reactivity	Very stable, unreactive under normal conditions	Electron-rich, vulnerable to attack by electrophiles

Types of isomers

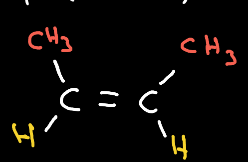


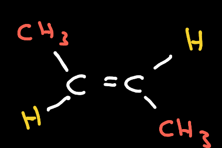
Stereoisomers: compounds same molecular and structural formuals but atoms or groups arranged differetly in 3D.

–Geometric Isomers

Difference: compound containing a C=C with atoms or groups attached at different positions.
Cause: 1– Restricted rotation about the carbon–carbon double bond (π bond).
2–There are two different groups on each of the carbon atoms in the C=C bond / double bond

Example: (But–2–ene)

① 

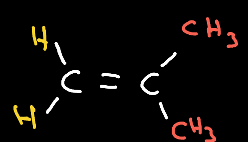
② 

Note: Alkanes can't have stereoisomers as they only have single bondeds

When Alkenes CANNOT Have Geometric Isomers

Condition: An alkene cannot have geometric isomers if at least one of the double–bonded carbons has two identical substituents (groups or atoms).

Example:



Cis and Trans naming system

Look for two identical groups or atoms, one attached to each carbon of the double bond.
–Cis (means "same side"): The two identical groups are on the SAME SIDE of the molecule.
–Trans (means "across"): The two identical groups are on OPPOSITE SIDES.

Cis and Trans naming ONLY works when both of the following are true:
–Each carbon of the double bond has two different groups attached (so rotation matters).
–There are two identical substituents, one on each carbon of the double bond, that can be directly compared.
If this condition is not met, the system fails.

E/Z naming system

Unlike cis/trans (which compares identical groups), E/Z uses a priority system. It compares the higher–priority group on each carbon of the double bond.

Process:
–For EACH carbon of the double bond, independently assign priority to its two attached groups using the Cahn–Ingold–Prelog (CIP) Rules.
–Identify the higher–priority group on Carbon 1 and the higher–priority group on Carbon 2.

Look at the spatial arrangement of these two high–priority groups:

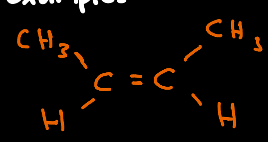
Z (from German zusammen, meaning "together"): The two high–priority groups are on the SAME SIDE.
E (from German entgegen, meaning "opposite"): The two high–priority groups are on OPPOSITE SIDES.

The Priority Rules

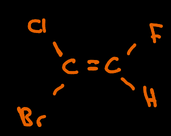
Rule 1: Atomic Number
Look at the atom directly attached to the double bond carbon.
Higher atomic number = higher priority.
Example: Br (35) > Cl (17) > S (16) > F (9) > O (8) > N (7) > C (6) > H (1)

Rule 2: If First Atoms Are Identical (Tie)
Move outward to the next set of atoms connected to those first atoms.
Compare their atomic numbers as a list, in descending order.
Example: $-CH_3$ vs $-CH_2CH_3$
First atoms: Both are C (tie).
Move out:
$-CH_3$ attaches to (H, H, H) → atomic numbers (1, 1, 1)
$-CH_2CH_3$ attaches to (C, H, H) → atomic numbers (6, 1, 1)
Compare: (6,1,1) vs (1,1,1) → 6 > 1
Result: $-CH_2CH_3$ has higher priority than $-CH_3$.

Examples

① 

Z-but-2-ene
OR
cis-but-2-ene

② 

E-1-bromo-1-chloro-2-floro ethene

③ 

E-pent-2-ene

Reactions

Introduction

Alkenes are a homologous series of unsaturated hydrocarbons characterized by having at least one carbon–carbon double bond (C=C). The presence of this π (pi) bond, which is electron-rich and relatively weak, makes alkenes much more reactive than alkanes. They primarily undergo **electrophilic addition addition reactions**, where the π bond breaks open and two new atoms or groups add to the carbons.

Addition reaction: reaction in which two molecules combine to form one molecule

Electrophile: chemical species that is electron-deficient and seeks to accept a pair of electrons to form a new bond. (electron lover)

Types of electrophiles:
1)Positively charged ions (cations like H^+, NO_2^+)
2)Molecules with a partial positive charge (δ^+, like the H in H–Br)
3)Molecules with an incomplete octete.g. (BF_3, carbocations)

Electrophiles with alkenes

Alkenes have an electron-rich π bond (a cloud of electrons above and below the C=C). Electrophiles are drawn to and attack this electron-rich area, starting the reaction.

Example: In the reaction of ethene with HBr:

HBr is polar: H is δ^+, Br is δ^-.
The δ^+ H acts as the electrophile: It is attracted to the π electrons of the double bond.
The H accepts the electron pair from the π bond, breaking the π bond and forming a new C-H bond.

Addition of Hydrogen (Hydrogenation)

–Reagents & Conditions: Hydrogen gas (H_2), Nickel (Ni) catalyst .
–Mechanism Type: addition/reduction (gaining hydrogen reduces the alkene).
–What Happens: The H-H bond breaks, and a hydrogen atom adds to each carbon of the double bond.
–Product: An alkane (saturated hydrocarbon).
–Equation Example:

$C_2H_4(g) + H_2(g) \rightarrow C_2H_6(g)$
Ethene → Ethane

Use: To harden vegetable oils (unsaturated fats) into margarine (saturated/partially saturated fats).
note: vegatible oils are partial hydrogenated so not all double bonds are broken

Addition of Halogens (Halogenation)

–Reagents: Bromine (Br_2) or Chlorine (Cl_2)
–Mechanism: Electrophilic addition.
–What Happens: The halogen molecule (X–X) is polarized and adds across the double bond.
–Product: A di-substituted halogenoalkane (dihaloalkane).
Equation Example:

$C_2H_4 + Br_2 \rightarrow CH_2Br{-}CH_2Br$
Ethene → 1,2-Dibromoethane (colorless)

Observable Test: Decolorization of brown liquid bromine(or orange aqueous bromine) . This is the standard test for unsaturation (C=C or C≡C bonds).
if added to alkanes the color will remain brown

Addition of Steam (Hydration)

–Reagents & Conditions: Steam ($H_2O(g)$), Concentrated phosphoric acid (H_3PO_4) .
–What Happens: This is an acid-catalyzed addition. The alkene reacts with water (H-OH).
–Product: An alcohol.

Equation Example:
$C_2H_4(g) + H_2O(g) \rightarrow CH_3CH_2OH(g)$

Ethene ⇌ Ethanol (reversible reaction)

Industrial Use: The primary industrial method for manufacturing ethanol (solvent, fuel).

Addition of Hydrogen Halides (HX)

–Reagents: Gaseous hydrogen halides (HCl, HBr, HI).
–Mechanism: Electrophilic addition.
–What Happens: The polar H–X bond adds across the double bond: H^+ (electrophile) adds first, then X^- (nucleophile).
–Product: A mono-substituted halogenoalkane.

Equation Example:
$C_2H_4 + HBr \rightarrow CH_3{-}CH_2Br$
Ethene → Bromoethane

Oxidation with Acidified Aqueous Potassium Manganate(VII)

–Reagents & Conditions: acidified potassium manganate(VII) solution ($KMnO_4 + H_2SO_4$).
–Mechanism Type: Oxidation & addition
–What Happens: The purple MnO_4^- ion oxidizes the C=C bond. The π bond breaks and OH groups add to each carbon.
–Observations: Purple color disappears (decolorizes).
–Product: A diol (a molecule with two -OH groups on adjacent carbons).
–Equation Example (Ethene):

$CH_2{=}CH_2 + [O] + H_2O \rightarrow CH_2OH{-}CH_2OH$
Ethene → Ethane-1,2-diol

Addition of Hydrogenalkanes Major and minor products

Important Note – for unsymmetrical alkenes (e.g. prop-2-ene) two alkanes produces are formed (major and minor)
Markovnikov's Rule: Hydrogen adds to the carbon with more hydrogens
Why? : adding H^+ to the carbon with more H's creates the more stable carbocation on the carbon with fewer H's.
–This stable carbocation intermediate then reacts with X^- to give the major product.
Therefore: The major product is the one where the halogen (X) is on the more substituted carbon as it is more stable. (tertiary > secondary > primary)
Primary Carbon (1°): A carbon atom bonded to only one other carbon
Example: $CH_3{-}CH_2{-}CH_3$ (the first C is bonded to one carbons).
Secondary Carbon (2°): A carbon atom bonded to two other carbon
Example: $CH_3{-}CH_2{-}CH_3$ (the second C is bonded to two carbons).
Tertiary Carbon (3°): A carbon atom bonded to threee other carbon
Example: $(CH_3)_3C{-}H$ (The central C is tertiary).

Explaining/Justifying Major and Minor Products:
Identify the intermediate carbocations.
Example: "Product A forms from a tertiary carbocation. Product B forms from a primary carbocation."
Compare their stability.
Example: "A tertiary carbocation is more stable than a primary carbocation."
State the conclusion.
Example: "Therefore, A is the major product and B is the minor product."

Polymerisation

Mechanism of addition

(halogens and hydrogen halids addition)

Step 1: Electrophilic Attack & Carbocation intermediate Formation:

The alkene's electron-rich π bond attacks the δ^+ atom in the polar molecule (H in H–Br or Br in Br–Br).

The H–Br or Br–Br bond breaks heterolytically (unevenly , both electrons go to one atoms).

H^+ (Br^+) adds to one carbon, forming a new C–H bond.

A carbocation forms on the other carbon and a bromide ion (Br^-).

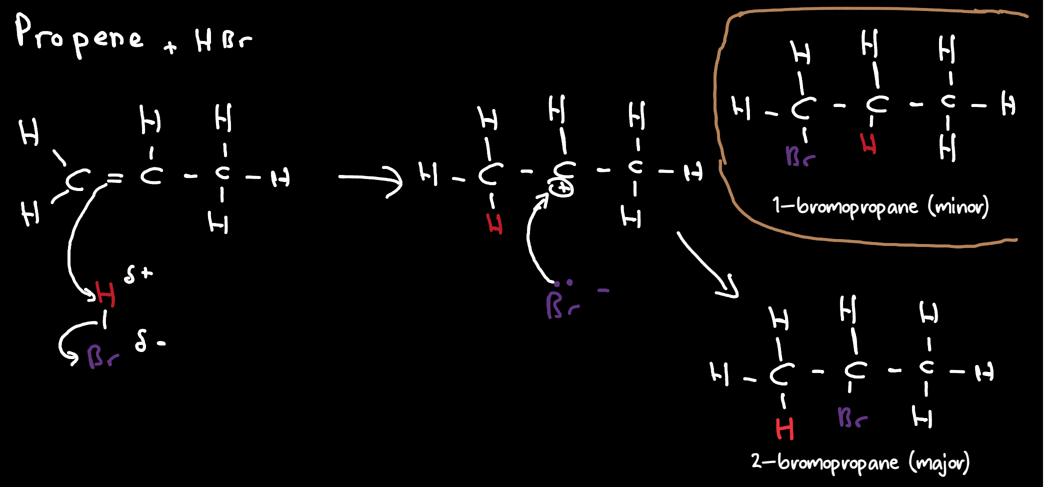
Step 2: Nucleophilic Attack by the Halide Ion

The carbocation (electron-deficient) is attacked by the nucleophilic Br^-.

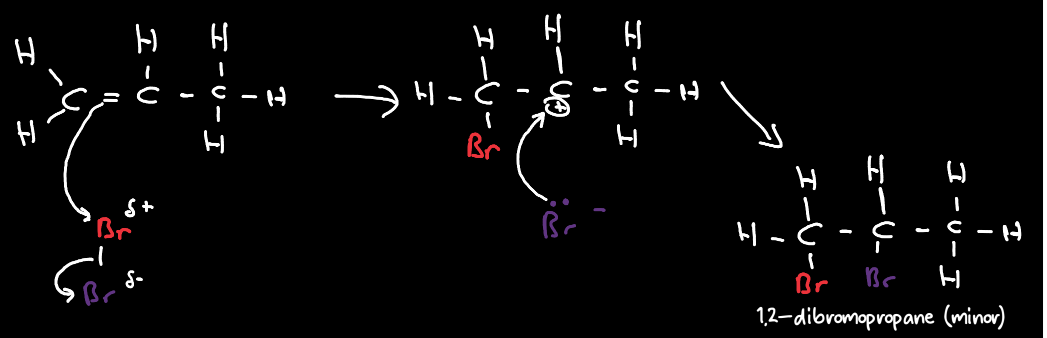
A new C–Br bond forms, yielding the final haloalkane.

note: in case hydrogen halids react with **asymmetrical** alkenes two products are formed (rule mentioned in the previous page)

Propene + HBr



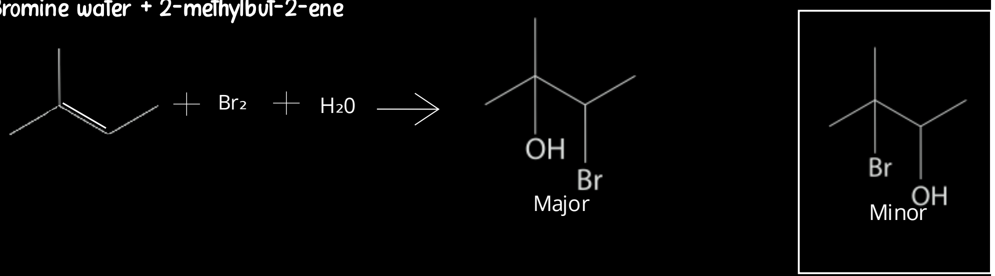
Propene + Br_2



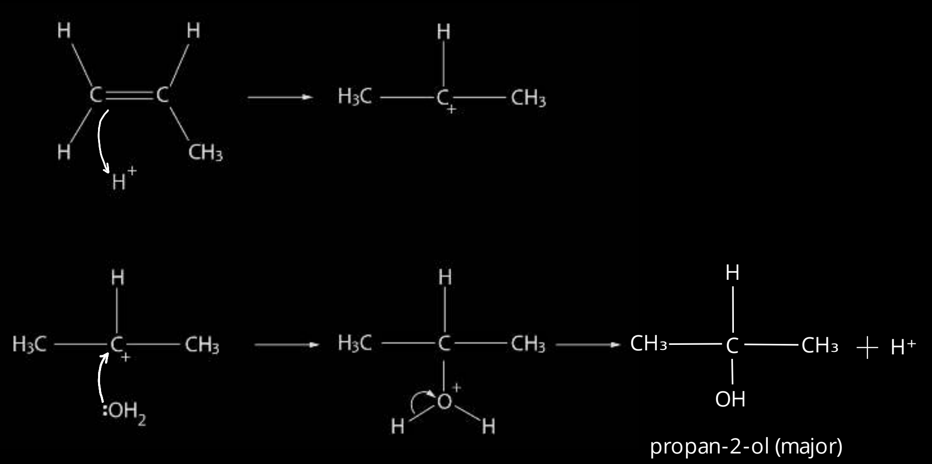
The double bond (π-bond) of the alkene repel electrons (distorts the electron cloud) in the bromine molecule / induces a dipole making the closer atoms slightly positive

Honorable mentions

Bromine water + 2-methylbut-2-ene



Steam + propene



Introduction

Definition: Polymerisation is a chemical reaction where many small molecules (monomers) join together to form a very large molecule (polymer).

For alkenes, this is specifically addition polymerisation.

–Reagents/Conditions: High pressure & temperature, often with a catalyst.
–Mechanism Type: Chain-growth addition polymerisation.
–What happens: The C=C double bond opens, and monomers link together by forming new single bonds.
–Key Feature: The polymer retains all the atoms from the monomer—no other product is formed.

The Naming System

Naming a polymer is straightforward once you know the monomer.

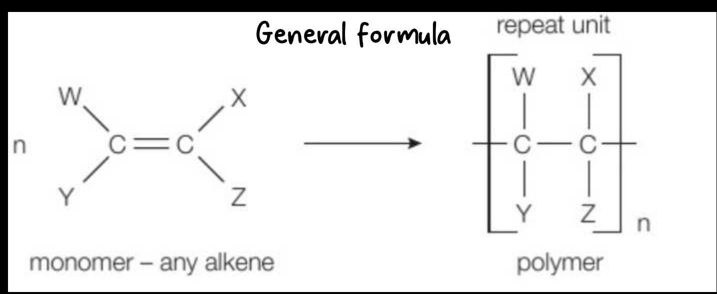
Rule: "Poly(monomer name)"

Simply put the name of the monomer in brackets after the prefix "poly-".

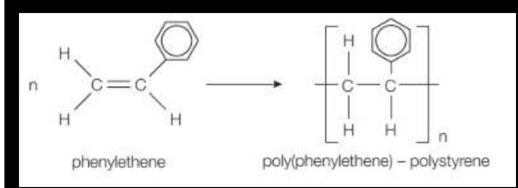
Addition vs Condensation

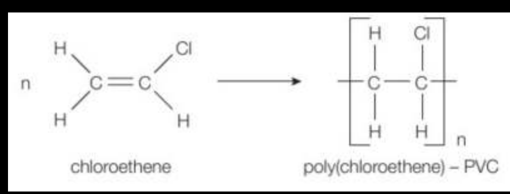
Addition Polymerisation:	Condensation Polymerisation:
–Monomers add together **without** losing any atoms. It's like linking train carriages together.	–Monomers join by releasing a small molecule (usually H_2O, HCl, or NH_3). Some material (water) is lost in the process.
–Needs unsaturated monomers .	–Needs two different functional groups on the monomers (e.g. –COOH & –OH, or –NH_2 & –COOH).
–Only one product (higher atom economy)	–Two products

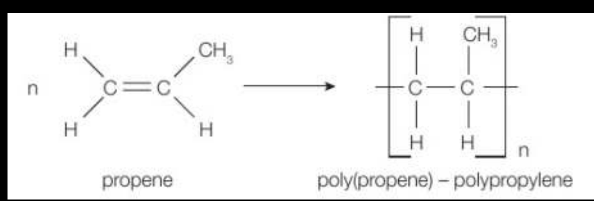
(further knowledge of condensation polymerization is not required)



Examples







Monomer vs polymers

- Mnemers are simple molecule That have weak intermolecular forces
- Polymers are macromolecules that have strong intermolecular forces

The Problem with Polymers

The widespread use of synthetic polymers (especially non-biodegradable plastics) has led to a major waste management crisis. These materials are durable, cheap, versatile, light and unreactive, but these same properties make them environmentally persistent when discarded.

Key Problems:

Persistence in Landfills: Most polymers are non-biodegradable and can remain in landfills for hundreds of years, taking up valuable space.

Environmental Pollution: Improper disposal leads to litter in terrestrial and marine environments, causing harm to wildlife through ingestion and entanglement.

Toxicity from Incineration: Burning polymers without proper controls releases toxic gases (e.g., sulfur dioxide, hydrogen chloride) and particulates, contributing to air pollution and health issues.

Greenhouse Gas Emissions: Incineration produces carbon dioxide (CO_2), a greenhouse gas, and some polymers release potent greenhouse gases like methane as they slowly break down anaerobically in landfills (decomposition).

Resource Depletion: Most polymers are made from non-renewable fossil fuels (crude oil and natural gas).

Incineration of Polymers

Incineration: A waste treatment process that involves the combustion of organic substances contained in waste materials, converting them into ash, flue gas, and heat.

Advantages of Incineration:

*Produces energy (heat/electricity) that can be used.
*Prevents polymers from taking up space in landfills.
Reduce risk of water/soil pollution

Disadvantages of Incineration:

*Can produce toxic substances (e.g., sulfur dioxide, dioxins, hydrogen chloride)
Particulates/soot may be released, causing respiratory problems.
*Produces CO_2, a greenhouse gas contributing to climate change.
Ash residue may contain toxic heavy metals.
High capital and operating costs for advanced emission-control systems.

Minimise Disposal of Polymers Problems

- *Help develop biodegradable polymers from plant material (polymers that can be re-used / recycled)
- *Remove toxic gases produced by incineration of polymers
- Develop processes to convert polymers back into feedstock (for use in chemical industry)
- Use of IR spectroscopy to separate polymers (for recycling)

Using Renewable Resources

Renewable Resource: A natural resource that can be replenished naturally with the passage of time (e.g., plants, crops) as opposed to finite fossil fuels.

"*": indicates more accurate answer, which are mostly mark scheme answers

Advantages:

*Renewable – can be replenished over time.
Often have a lower carbon footprint than fossil-fuel-based polymers.
Can support sustainable agriculture/forestry.

Disadvantages:

*Loss of food resources if food crops (e.g., corn, sugarcane) are diverted for plastic production.
*Require land to grow, reducing land available for food production (food vs. fuel debate).
May involve large-scale monoculture, impacting biodiversity.
Can have high water and fertilizer requirement.

Developing Biodegradable Polymers

Biodegradable Polymer: A polymer that can be broken down by the natural action of microorganisms/decomposers (such as bacteria and fungi) into water, carbon dioxide, methane, and biomass.

Advantages:

*Take less time to break down compared to most traditional plastics.
*Reduce waste going to landfill.
*Do not require incineration.
Reduce pollution, litter, and harm to wildlife.
Break down into non-harmful products (e.g., CO_2, H_2O, biomass).
Often come from a renewable resource/sustainable.

Disadvantages:

*loss of food resources (in case of using plants)
Can be more expensive to produce than conventional polymers.
*Can not be recyled, incinerated etc...
Material properties (e.g., strength, durability) may be inferior to traditional plastics.

Recycling of Polymere

Definition: The process of collecting, sorting, processing, and converting waste polymer materials into new products or raw materials.

Advantages of Recycling:

*Conserves Resources
*Reduces Energy Consumption: recycling uses less energy
*Reduces Landfill Waste
Lowers Carbon Footprint: Reduces greenhouse gas emissions (e.g., CO_2)
Less toxic substances released

Disadvantages / Challenges of Recycling:

*Energy is still consumed to make the new product
*Polymers need to be stored (which is expensive)
Limited Recyclability: not all polymers can be recycled (they need to be separated to be recycled)
Degradation: Polymers can degrade with each recycling cycle due to chain scission, limiting the number of times they can be effectively recycled.